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Cell Types Populating the Medial Temporal Lobe and Amygdala

Table of Contents

Introduction/Background

The medial temporal lobe and amygdala do not contain one simple set of cells. They contain many neuron and glial cell types, and these differ by region, layer, transmitter, genes, shape, connections, and developmental origin. (LLM Memory)

When people ask for "all the cell types" in the medial temporal lobe and amygdala, the main challenge is that these are large, mixed systems rather than single nuclei or one kind of cortex. The medial temporal lobe usually includes the hippocampal formation, entorhinal cortex, perirhinal and parahippocampal regions, and closely linked amygdala-related areas, while the amygdala itself contains several nuclei with distinct cell makeup. Across these structures, the broad cell classes are excitatory projection neurons, inhibitory interneurons, modulatory input fibers from other brain regions, glial cells such as astrocytes, oligodendrocytes and their precursors, microglia, vascular cells, ependymal-like barrier cells in some zones, and in some subregions small pools of immature or adult-born neurons. (Model-Generated)

A full answer also needs to separate different ways of defining a "cell type." In these regions, cell types can be grouped by neurotransmitter use, for example glutamatergic versus GABAergic neurons; by anatomy, such as pyramidal, granule, semilunar, stellate, spiny projection, or chandelier-like cells; by molecular markers and transcriptomic state; by where they are born in development; and by what circuits they join. Because of this, no single final list is fully complete or fixed, and modern single-cell RNA sequencing often splits older broad classes into many finer subclasses. (Model-Generated)

For that reason, the best way to answer the user's question is to move from broad shared classes to region-specific populations. The sections below first outline the main cellular makeup of medial temporal lobe structures, then the major classes in amygdala nuclei, then finer neuronal subtypes, developmental-origin groups, non-neuronal populations, and newer atlas-style cell diversity maps that reveal still more specialized classes. (Model-Generated)

Medial temporal lobe structures and broad cellular makeup

The medial temporal lobe includes the hippocampal formation, entorhinal cortex, surrounding parahippocampal and perirhinal cortices, and the amygdala. Across these areas, the broad cell makeup is mainly excitatory principal neurons plus inhibitory interneurons, with cell organization changing by subregion and nucleus. (9 sources)

The medial temporal lobe is best understood as a set of linked structures rather than one uniform tissue block. It includes the hippocampal formation, entorhinal cortex, perirhinal and parahippocampal cortices, and the amygdala, with some descriptions also emphasizing nearby olfactory-linked medial temporal cortical areas such as piriform cortex. (Preston et al., 2002) (Heckers, 2000) (Juran et al., 2016) More fine-grained anatomical descriptions break the hippocampal formation into dentate gyrus, CA1-CA4, and subiculum, and place the amygdala as a multi-nuclear complex in the medial temporal pole that partly overlaps the hippocampal head. (Nolan et al., 2020) (Kiernan, 2012) In functional and imaging work, these medial temporal lobe systems are often seen as longitudinally organized networks that span parts of the amygdala, hippocampus, and entorhinal cortex, again showing that the same broad territory contains several distinct cellular environments rather than one single cell set. (Ruiz-Rizzo et al., 2019) (Yousuf et al., 2021) (Poggi et al., 2024)

At the broadest cellular level, the hippocampal region is dominated by principal excitatory neurons and a smaller set of inhibitory interneurons. In the human hippocampal region, neurons are arranged in a single main cellular layer called the pyramidal cell layer, where most pyramidal neurons are glutamatergic and the smaller nonpyramidal population is GABAergic. (Heckers, 2000) That broad rule extends across medial temporal lobe structures in modified form: hippocampal CA fields are rich in pyramidal excitatory cells, dentate gyrus is built around excitatory granule cells, subiculum contains projection neurons, and adjacent cortices such as entorhinal, perirhinal, and parahippocampal cortex contain cortical excitatory projection neurons mixed with diverse GABAergic interneurons. (Nolan et al., 2020) (Preston et al., 2002) The amygdala adds another level of variation because it is made of several nuclei rather than one cortex-like sheet, and these nuclei are commonly grouped into deep laterobasal nuclei, superficial cortical-like nuclei, and centromedial nuclei, each with its own balance of projection neurons and inhibitory cells. (Nolan et al., 2020) (Yang et al., 2017) So, for the user's question, the broad answer for the medial temporal lobe is: excitatory principal neurons of several kinds, inhibitory interneurons of several kinds, and then region-specific supporting non-neuronal cells discussed later; but the exact mix depends strongly on whether one is in dentate gyrus, CA fields, subiculum, entorhinal and parahippocampal cortex, or one of the amygdala nuclei.

Evidence

(Preston et al., 2002)

Different functions for different medial temporal lobe structures?

> *"The MTL is comprised of multiple structures including the hippocampal formation (the dentate gyrus, CA fields, and the subiculum), the amygdala, entorhinal cortex, and surrounding perirhinal and parahippocampal cortices."*

(Heckers, 2000)

Neural models of schizophrenia

> *"The medial temporal lobe contains the amygdala, the hippocampal region, and superficial cortical areas that cover the hippocampal region and form the parahippocampal gyrus (PHG). The hippocampal region can be subdivided into three subregions: the dentate gyrus (DG), the cornu ammonis (CA) sectors, and the subiculum (Sub). The neurons of the human hippocampal region are arranged in one cellular layer, the pyramidal cell layer. Most pyramidal cell layer neurons are glutamatergic whereas the small contingent of nonpyramidal cells are GABAergic."*

(Juran et al., 2016)

Unilateral Resection of the Anterior Medial Temporal Lobe Impairs Odor Identification and Valence Perception

> *"Several cortical areas located within the medial TL, among others the piriform cortex, the amygdala, the entorhinal cortex as well as the hippocampal formation, have been linked to olfactory processing due to their close anatomical connectivity to the olfactory receptor neurons (Room et al., 1984)(Carmichael et al., 1994)."*

(Nolan et al., 2020)

Hippocampal and Amygdalar Volume Changes in Major Depressive Disorder: A Targeted Review and Focus on Stress

> *"The hippocampus proper consists of the allocortical cornu ammonis subfields 1-4 (CA1-4), and the hippocampal formation consists of the dentate gyrus (DG) medially along its transverse axis, and the subiculum inferiorly"*

".amygdala nuclei are commonly categorised into three groups: the deep laterobasal amygdala containing the lateral (LA) and basal nuclei; the superficial cortical-like nuclei; and centromedial amygdala containing the central (CE) and medial nuclei. (Yang et al., 2017)"

(Kiernan, 2012)

Anatomy of the Temporal Lobe

"The adult amygdala is a group of several nuclei located in the medial part of the temporal pole, anterior to and partly overlapping the hippocampal head."

(Ruiz-Rizzo et al., 2019)

Human subsystems of medial temporal lobes extend locally to amygdala nuclei and globally to an allostatic-interoceptive system

"These subsystems consistently covered parts of the amygdala, hippocampus, and entorhinal cortex, and showed a discrete longitudinal organization along the medial temporal lobes."

(Yousuf et al., 2021)

Functional coupling between CA3 and laterobasal amygdala supports schema dependent memory formation

"Within the medial temporal lobe, semantic congruence was associated with hemodynamic activity in the subiculum, CA1, CA3 and dentate gyrus, as well as the entorhinal cortex and laterobasal amygdala."

(Poggi et al., 2024)

Pathophysiology in cortico-amygdala circuits and excessive aversion processing: the role of oligodendrocytes and myelination

"he medial temporal lobe (MTL), including amygdala and hippocampus"

(Yang et al., 2017)

From Structure to Behavior in Basolateral Amygdala-Hippocampus Circuits

"Emotion influences various cognitive processes, including learning and memory. The amygdala is specialized for input and processing of emotion, while the hippocampus is essential for declarative or episodic memory. During emotional reactions, these two brain regions interact to translate the emotion into particular outcomes. Here, we briefly introduce the anatomy and functions of amygdala and hippocampus, and then present behavioral, electrophysiological, optogenetic and biochemical evidence from recent studies to illustrate how amygdala and hippocampus work synergistically to form long-term memory. With recent technological advances, the causal investigations of specific neural circuit between amygdala and hippocampus will help us understand the brain mechanisms of emotion-regulated memories and improve clinical treatment of emotion-associated memory disorders in patients."

Amygdala organization and principal cellular classes

The amygdala is not one cell population but a set of more than a dozen nuclei. Its main cell classes change by nuclear group: cortical-like and basolateral parts are mostly glutamatergic projection neurons plus GABAergic interneurons, while central and much of the extended amygdala are mainly GABAergic output neurons. (14 sources)

The amygdala sits in the medial temporal lobe as a cytoarchitectonically mixed complex of at least 13 nuclei and cortical areas, many of which are further subdivided rather than being single uniform structures. It is commonly organized into a deep or basolateral group, a superficial or cortical-like group, a centromedial group, and several additional nuclei such as the anterior amygdaloid area, amygdalohippocampal area, and intercalated cell masses; some authors also include a rostro-medial “extended amygdala” that links parts of the amygdala with bed nucleus of the stria terminalis territory. (Pitkanen et al., 1994) (Ignacio et al., 2014) (Nardelli et al., 2024) (Pitkanen et al., 1997) (Porcaro et al., 2023) (Kinkead et al., 2023)

For cell types, the key first split is between cortical-like amygdala regions and striatum-like or extended amygdala regions. The basolateral amygdala, especially the lateral, basal, and accessory basal nuclei, is repeatedly described as a cortical-like structure built mainly from glutamatergic projection neurons of pyramidal or pyramidal-like type, together with a smaller but important population of local GABAergic interneurons; estimates in the cited work put excitatory neurons at about 70-80% of neurons in this division. (Veinante et al., 2013) (Polepalli et al., 2020) (Perumal et al., 2021) (Ignacio et al., 2014) (Cutsuridis et al., 2017)

The superficial cortical amygdala also fits this cortex-like pattern, while the central nucleus differs sharply. In the central amygdala, the majority of neurons are thought to be GABAergic, and work comparing amygdala divisions notes that the extended amygdalar nuclei contain mainly GABAergic spiny projection neurons, more like the striatum than cortex. (Ignacio et al., 2014) (Loonen et al., 2016) (McDonald et al., 2016)

The medial amygdala is more mixed. Recent summaries describe principal neurons as exclusively glutamatergic in the BLA, cortical amygdala, and basomedial amygdala, exclusively GABAergic in the central amygdala, and predominantly GABAergic in the medial amygdala and bed nucleus of the stria terminalis, while also noting smaller glutamatergic pyramidal populations in parts of the medial and extended amygdala in rodents. (Raudales et al., 2024) (Huilgol et al., 2016)

A separate principal class that matters for amygdala circuit control is the intercalated cell masses. These are small clusters of densely packed GABAergic neurons positioned between major nuclei and are often treated as a distinct amygdalar population rather than just another interneuron set. (Veinante et al., 2013) (Ignacio et al., 2014)

So if the user wants the main amygdala cell classes before finer subtype lists, the short answer is: glutamatergic pyramidal or pyramidal-like projection neurons in basolateral and other cortical-like nuclei; GABAergic interneurons within those cortical-like nuclei; predominantly or exclusively GABAergic projection neurons in central and extended amygdala territories; mixed but mostly GABAergic principal populations in medial/extended groups; and specialized GABAergic intercalated cell clusters between nuclei. Neuromodulatory inputs such as cholinergic afferents are also part of the cellular and synaptic environment, even though their cell bodies often lie outside the amygdala itself. (Cutsuridis et al., 2017)

Evidence

(Pitkanen et al., 1994)

The distribution of GABAergic cells, fibers, and terminals in the monkey amygdaloid complex: an immunohistochemical and in situ hybridization study

> *"The monkey amygdaloid complex is a cytoarchitectonically heterogeneous structure that is composed of at least 13 different nuclei and cortical areas, many of which are subdivided into two or more regions."*

(Ignacio et al., 2014)

Effects of Acute Prenatal Exposure to Ethanol on microRNA Expression are Ameliorated by Social Enrichment

> *"At the cellular level, the amygdala is composed of a group of 13 sub-nuclei located in the medial temporal lobe (Price, 2003). These nuclei may be divided into four subdivisions (Sah et al., 2003): (Ethen et al., 2009) basolateral (which includes the lateral, basolateral, and basomedial nuclei), (May et al., 2009) cortical like (including nucleus of the lateral olfactory tract, bed nucleus of the accessory olfactory tract, the cortical nucleus, and the periamygdaloid cortex), (3) centromedial (central and medial nuclei, and the amygdaloid part of the bed nucleus of stria terminalis), and (4) other (which includes anterior amygdala area, the amygdalo-hippocampal area, and the intercalated nuclei)"*

> *".In the basolateral group, approximately 70% of neurons are thought to be glutamatergic (pyramidal, spiny, or class I neurons). This division also contains interneurons such as GABAergic nonspiny stellate cells of the cortex (called S cells, stellate, or class II neurons). In contrast, within the central nucleus, the majority of cells are thought to be GABAergic."*

(Nardelli et al., 2024)

Pain in Parkinson's disease: a neuroanatomy-based approach

> *"The amygdala is a bilateral structure located in the medial temporal lobe"*

> *". Anatomically the amygdala is composed of three major nuclear groups 198 : the deep or basolateral group, which contains the lateral nucleus, the basal nucleus, and the accessory basal nucleus; the superficial or cortical-like group, which contains the cortical nuclei and the nucleus of the lateral olfactory tract; the centromedial group, which contains the medial and central nuclei. To this canonical classification, other amygdaloid nuclei must be added, such as the anterior amygdaloid area, the amygdalohippocampal area, and the intercalated cells. 199 In addition, a rostro-medial extension of the centromedian amygdala into an area known as extended amygdala has been proposed. 200"*

(Pitkanen et al., 1997)

Organization of intra-amygdaloid circuitries in the rat: an emerging framework for understanding functions of the amygdala.

"The amygdala is located in the medial aspects of the temporal lobe. In spite of the fact that the amygdala has been implicated in a variety of functions, ranging from attention to memory to emotion, it has not attracted neuroscientists to the same extent as its laminated neighbours, in particular the hippocampus and surrounding cortex. However, recently, principles of information processing within the amygdala, particularly in the rat, have begun to emerge from anatomical, physiological and behavioral studies. These findings suggest that after the stimulus enters the amygdala, the highly organized intra-amygdaloid circuitries provide a pathway by which the representation of a stimulus becomes distributed in parallel to various amygdaloid nuclei. As a consequence, the stimulus representation may become modulated by different functional systems, such as those mediating memories from past experience or knowledge about ongoing homeostatic states. The amygdaloid output nuclei, especially the central nucleus, receive convergent information from several other amygdaloid regions and generate behavioral responses that presumably reflect the sum of neuronal activity produced by different amygdaloid nuclei."

(Porcaro et al., 2023)

Seeking the Amygdala: Novel Use of Diffusion Tensor Imaging to Delineate the Basolateral Amygdala

"The amygdala itself is a constellation of nuclei and cell types that are remarkably conserved among mammalian species, including rat, monkey and human (Chareyron et al., 2011). The amygdala is primarily composed of the basolateral (BLA) amygdala (lateral, basal, and accessory basal nuclei), and the central, medial and cortical nuclei, which have been termed the amygdaloid complex (Davis et al., 2000)(Pitkänen et al., 1997)."

(Kinkead et al., 2023)

Estrogens, age, and, neonatal stress: panic disorders and novel views on the contribution of non-medullary structures to respiratory control and CO2 responses

"Within the medial temporal lobes, the amygdalar complex is responsible for perception and processing of stimuli; it also initiates and terminates emotional reactions (Marek et al., 2013). Briefly, this complex is composed of three main structures: the medial amygdala (MeA), the central amygdala (CeA) and the basolateral amygdala (BLA) (Figure 2)."

(Veinante et al., 2013)

The amygdala between sensation and affect: a role in pain

"Its main nuclei, the lateral and the basolateral nuclei are known as the basolateral amygdala (BLA) and mostly contain projection neurons of pyramidal type, synthesizing glutamate"

". In addition to the four groups, a few nuclei of the amygdala remain unclassified, among them the intercalated cell masses which are small clusters of densely packed GABAergic neurons (Palomares-Castillo et al., 2012)."

(Polepalli et al., 2020)

Diversity of interneurons in the lateral and basal amygdala

"It is a cortical-like structure and contains glutamatergic pyramidal neurons and GABAergic interneurons."

(Perumal et al., 2021)

Inhibitory Circuits in the Basolateral Amygdala in Aversive Learning and Memory

"The BLA is a cortical like structure, and similar to other cortical regions, local microcircuits are formed by excitatory glutamatergic neurons that make [illegible] 80%of total neuronal population and GABAergic interneurons (Sah et al., 2003)(Tovote et al., 2015)."

(Cutsuridis et al., 2017)

Editorial: Memory Processes in Medial Temporal Lobe: Experimental, Theoretical and Computational Approaches

"Principal (excitatory) cells"

"inhibitory interneurons"

"neuromodulators such as acetylcholine"

(Loonen et al., 2016)

Circuits Regulating Pleasure and Happiness: The Evolution of the Amygdalar-Hippocampal-Habenular Connectivity in Vertebrates

"Both the cortical amygdalar nuclei and the basolateral amygdalar nuclear complex, which is located deeper within the amygdaloid complex, have cortex-like cell types (McDonald et al., 2016). In contrast, the so called "extended amygdalar nuclei" contain predominantly GABAergic spiny projection neurons, like the striatum (McDonald et al., 2016)."

(McDonald et al., 2016)

Functional Neuroanatomy of Amygdalohippocampal Interconnections and Their Role in Learning and Memory

> *"The amygdalar nuclear complex and hippocampal/parahippocampal region are key components of the limbic system that play a critical role in emotional learning and memory. This Review discusses what is currently known about the neuroanatomy and neurotransmitters involved in amygdalo-hippocampal interconnections, their functional roles in learning and memory, and their involvement in mnemonic dysfunctions associated with neuropsychiatric and neurological diseases. Tract tracing studies have shown that the interconnections between discrete amygdalar nuclei and distinct layers of individual hippocampal/parahippocampal regions are robust and complex. Although it is well established that glutamatergic pyramidal cells in the amygdala and hippocampal region are the major players mediating interconnections between these regions, recent studies suggest that long-range GABAergic projection neurons are also involved. Whereas neuroanatomical studies indicate that the amygdala only has direct interconnections with the ventral hippocampal region, electrophysiological studies and behavioral studies investigating fear conditioning and extinction, as well as amygdalar modulation of hippocampal-dependent mnemonic functions, suggest that the amygdala interacts with dorsal hippocampal regions via relays in the parahippocampal cortices. Possible pathways for these indirect interconnections, based on evidence from previous tract tracing studies, are discussed in this Review. Finally, memory disorders associated with dysfunction or damage to the amygdala, hippocampal region, and/or their interconnections are discussed in relation to Alzheimer's disease, posttraumatic stress disorder (PTSD), and temporal lobe epilepsy. © 2016 Wiley Periodicals, Inc."*

(Raudales et al., 2024)

[Specific and comprehensive genetic targeting reveals brain-wide distribution and synaptic input patterns of GABAergic axo-axonic interneurons]

> *"The amygdaloid complex includes over a dozen nuclei and can be segregated into five groups (Beyeler and Dabrowska, 2020): (1) the BLA divided into a dorsal section (lateral amygdala, LA) and basal section (basal amygdala, BA), (2) the basomedial amygdala (BMA), (3) the central amygdala (CeA) further splits into medial, lateral, and central sections (CeM, CeL, and CeC), (4) the medial amygdala (MeA), and (5) the cortical amygdala (CoA)"*

> *".the former includes BLA, CoA, BMA, and MeA, while the latter includes CeA and BST.Within the amygdala nuclei, PNs are exclusively glutamatergic in BLA, CoA, BMA, exclusively GABAergic in CeA, and predominantly GABAergic in MeA and BST.In rodents, there is also a population of glutamatergic pyramidal neurons (GLU PNs, derived from third ventricle neuroepithelium) that populates the BST, MeA, and hypothalamus (García-Moreno et al., 2010)(Huilgol et al., 2016)."*

(Huilgol et al., 2016)

[Cell migration in the developing rodent olfactory system]

> *"The components of the nervous system are assembled in development by the process of cell migration. Although the principles of cell migration are conserved throughout the brain, different subsystems may predominantly utilize specific migratory mechanisms, or may display unusual features during migration. Examining these subsystems offers not only the potential for insights into the development of the system, but may also help in understanding disorders arising from aberrant cell migration. The olfactory system is an ancient sensory circuit that is essential for the survival and reproduction of a species. The organization of this circuit displays many evolutionarily conserved features in vertebrates, including molecular mechanisms and complex migratory pathways. In this review, we describe the elaborate migrations that populate each component of the olfactory system in rodents and compare them with those described in the well-studied neocortex. Understanding how the components of the olfactory system are assembled will not only shed light on the etiology of olfactory and sexual disorders, but will also offer insights into how conserved migratory mechanisms may have shaped the evolution of the brain."*

Basolateral amygdala neuronal subtypes

In the basolateral amygdala, the main split is between glutamatergic principal neurons and a smaller but very diverse set of GABAergic interneurons. The interneuron side includes several recurring classes that look much like cortical inhibitory cell types, such as PV, SOM, CCK, VIP/calretinin, neurogliaform, axo-axonic, and some GABAergic projection cells. (5 sources)

- **Glutamatergic principal neurons**
- The main excitatory cells in the basolateral amygdala are **pyramidal-like projection neurons** with **spiny dendrites** that use **glutamate**. This is the dominant neuronal class in the basolateral complex. (McDonald et al., 2012) (Woodruff et al., 2007) (Unal et al., 2020)
- In broad numerical terms, the BLA is described as about **75% glutamatergic principal neurons**, with the rest mostly inhibitory interneurons. (Unal et al., 2020)
- Even within principal cells, morphology varies: in the lateral and basolateral nuclei, **some small neurons have dendritic fields restricted to limited sensory terminal zones**, while **larger neurons have more extensive dendritic fields**. This suggests that principal-cell-like populations are not all the same in input

sampling or integration. (Nishijo et al., 1988)

- **GABAergic interneurons as a broad class**
- The remaining roughly **25%** of BLA neurons form a **neurochemically and electrophysiologically heterogeneous** inhibitory population. (Unal et al., 2020)
- Older and notable reviews describe **four main interneuron populations** in the BLA, defined by marker expression: **parvalbumin (PV)**, **somatostatin (SOM)**, **cholecystokinin (CCK)**, and **calretinin/VIP-related** groups. (McDonald et al., 2012) (Woodruff et al., 2007)
- **PV-related interneurons**
- One major subtype is **PV+/CB+ interneurons**. (McDonald et al., 2012)
- More detailed cell-type mapping indicates that PV cells include at least two important classes:
- **PV basket cells**, estimated at about **17-20%** of GABAergic cells in the lateral and basal nuclei. (Vereczki et al., 2021)
- **Axo-axonic cells** (chandelier-like cells), estimated at about **5.5-6%**. (Vereczki et al., 2021)
- PV interneurons are one of the clearest parallels between BLA and neocortical inhibitory organization. (Unal et al., 2020)
- **SOM-related interneurons**
- Another major subtype is **SOM+/CB+ interneurons**. (McDonald et al., 2012)
- In newer classifications, these are largely **dendrite-targeting inhibitory cells expressing somatostatin**, making up about **10-16%** of GABAergic neurons. (Vereczki et al., 2021)
- SOM interneurons, like PV cells, are a notable and recurring BLA inhibitory class. (Woodruff et al., 2007) (Unal et al., 2020)
- **CCK-related interneurons**
- McDonald and colleagues describe **large multipolar CCK+ neurons**, often also **CB+**, as a distinct BLA interneuron group. (McDonald et al., 2012)
- More fine-grained work separates out **CCK-expressing basket cells**, estimated at about **7-9%** of GABAergic cells. (Vereczki et al., 2021)
- **Calretinin/VIP-related interneurons**
- A fourth classic group consists of **small bipolar and bitufted interneurons** with strong colocalization of **calretinin, CCK, and VIP**. (McDonald et al., 2012)
- In more recent BLA cell counts, **VIP- and/or calretinin-expressing interneuron-selective interneurons** are estimated to be the **largest single GABAergic fraction**, about **29-38%**. (Vereczki et al., 2021)
- **Neurogliaform and related inhibitory cells**
- A substantial BLA inhibitory class is formed by **NPY-containing neurogliaform cells**, estimated at about **14-15%** of GABAergic neurons in the lateral and basal nuclei. (Vereczki et al., 2021)
- **GABAergic projection neurons**
- Not all inhibitory neurons in the BLA are purely local interneurons. A smaller group of **GABAergic projection neurons** expressing **somatostatin and neuronal nitric oxide synthase** has been estimated at about **5.5-8%**

of GABAergic cells. (Vereczki et al., 2021)

- **Bottom line for the user's "what cell types are there?" question**
- If we list the main neuronal cell types currently described for the basolateral amygdala, they include: **glutamatergic pyramidal-like principal neurons**; **PV interneurons** including **basket** and **axo-axonic** cells; **SOM dendrite-targeting interneurons**; **CCK interneurons** including **CCK basket cells**; **VIP/calretinin interneuron-selective cells**; **NPY neurogliaform cells**; and **GABAergic projection neurons**. Together, these studies suggest that the LA and BA contain most of the major inhibitory neuron classes seen in other cortical regions. (McDonald et al., 2012) (Woodruff et al., 2007) (Vereczki et al., 2021)

Evidence

(McDonald et al., 2012)

Subpopulations of somatostatin-immunoreactive non-pyramidal neurons in the amygdala and adjacent external capsule project to the basal forebrain: evidence for the existence of GABAergic projection neurons in the cortical nuclei and basolateral nuclear complex

> *"The principal neurons in the CBL are pyramidal-like projection neurons with spiny dendrites that utilize glutamate as an excitatory neurotransmitter, whereas most non-pyramidal neurons in the CBL are spine-sparse interneurons that utilize GABA as an inhibitory neurotransmitter (McDonald, 1982)(McDonald, 1985)(McDonald, , 1992a(McDonald, ,b, 1996(McDonald, , 2003(Millhouse et al., 1983)(Fuller et al., 1987)(Carlsen et al., 1988)(McDonald et al., 1993). Dual-labeling immunohistochemical studies in the basolateral amygdala suggest that the CBL contains at least four distinct subpopulations of GABAergic non-pyramidal neurons that can be distinguished on the basis of their content of calcium-binding proteins and peptides. These subpopulations are: (1) PV+/CB+ neurons, (2) SOM+/CB+ neurons, (3) large multipolar cholecystokinin+ neurons that are often CB+, and (4) small bipolar and bitufted interneurons that exhibit extensive colocalization of calretinin, cholecystokinin, and vasoactive intestinal peptide (Kemppainen and Pitkänen, 2000;McDonald and Betette, 2001;Mascagni, 2001, 2002;"*

(Woodruff et al., 2007)

Networks of Parvalbumin-Positive Interneurons in the Basolateral Amygdala

> *"two main types of neuron have been described: glutamatergic principal neurons and GABAergic interneurons (McDonald, 1992; (Smith et al., 1998)"*

> *"Four populations of interneurons have been described in the BLA: those expressing parvalbumin (McDonald, 1992;Mc-Donald and Betette, 2001), those expressing somatostatin (Mc-Donald and Mascagni, 2002), those expressing cholecystokinin"*

(Unal et al., 2020)

Low-threshold spiking interneurons perform feedback inhibition in the lateral amygdala

> *"The BLA is a collection of smaller cortex-like nuclei and is composed ~ 75% of glutamatergic principal neurons"*

> *"The remaining population (~ 25%) constitutes a neurochemically and electrophysiologically heterogeneous set of interneurons"*

> *"The most salient parallels between BLA and other cortical regions with respect to their interneurons exist with respect to parvalbumin (PV) and somatostatin (SOM) positive interneurons."*

(Nishijo et al., 1988)

Topographic distribution of modality-specific amygdalar neurons in alert monkey

> *"At least within the lateral and basolateral nuclei, some small cells have dendritic fields that are confined to one or another of these limited sensory terminal areas (Millhouse and de Olmos, 1983). In the same areas, larger neurons have extensive dendritic fields (Millhouse and de Olmos, 1983)."*

(Vereczki et al., 2021)

Total Number and Ratio of GABAergic Neuron Types in the Mouse Lateral and Basal Amygdala

> *"we estimated that the following cell types together compose the vast majority of GABAergic cells in the LA and BA: axo-axonic cells (5.5%-6%), basket cells expressing parvalbumin (17%-20%) or cholecystokinin (7%-9%), dendrite-targeting inhibitory cells expressing somatostatin (10%-16%), NPY-containing neurogliaform cells (14%-15%), VIP and/or calretinin-expressing interneuron-selective interneurons (29%-38%), and GABAergic projection neurons expressing somatostatin and neuronal nitric oxide synthase (5.5%-8%). Our results show that these amygdalar nuclei contain all major GABAergic neuron types as found in other cortical regions."*

Medial and extended amygdala developmental-origin cell populations

A full cell-type answer for the medial and extended amygdala also has to include where cells come from in development. These regions are mosaics made from several embryonic sources, so their mature cell populations mix different transmitter types and lineage-defined neuron groups. (5 sources)

The medial amygdala and the nearby extended amygdala are especially important examples of why a single simple cell list is not enough. Developmental studies describe these regions as built from multiple embryonic domains rather than one local germinal source. Across amygdalar subdivisions, cells can arise from dorsal striatal, ventral striatal, pallidal, and preoptic embryonic domains, and they reach the amygdala by both radial and tangential migration. (Vicario et al., 2014) More recent work generalizes this point by describing several amygdalar regions as having a heterogeneous composition that includes pallial and subpallial parts, plus hypothalamic and other tangentially migrated subpallial cell types. (Garcia-Calero et al., 2020)

For the **medial amygdala** in particular, the cell populations are described as overlapping strongly with those of the **bed nucleus of the stria terminalis medial division (BSTM)** and related extended amygdala territories. These shared populations include cells of pallidal, preoptic, hypothalamic, and prethalamic eminence origin, and the medial amygdala also contains a smaller **Lhx9-positive** population from the **ventral pallium**. (Vicario et al., 2016) In another summary, the rodent medial amygdala is described as a mosaic made mainly of **GABAergic subpallial cells**, but complemented by **glutamatergic neuron types** that come from outside sources such as the **ventral pallium, SPV, and EmT**. (Gerlach et al., 2021)

This developmental view helps explain why the medial and extended amygdala contain mixed principal populations instead of a single uniform neuron class. In practical terms, the user should think of these regions as containing at least: **subpallial GABAergic lineage-derived neurons** from pallidal, preoptic, and striatal-related sources; **hypothalamic-related cell populations**; and smaller sets of **extrinsic glutamatergic neurons** from pallial or related sources. (Vicario et al., 2014) (Vicario et al., 2016) (Garcia-Calero et al., 2020) (Gerlach et al., 2021)

More fine-grained lineage studies further split these medial extended amygdala populations. For example, **TOH-domain-derived** double-labeled cells populate ventral parts of the **medial extended amygdala**, extending from **BSTM3 into part of the medial amygdala**, while **SPV core-derived Sim1 cells** form a distinct BSTM stripe, scatter through other medial extended amygdala areas, reach the arcopallium, and a subset also enters the **central extended amygdala**, especially its lateral part. (Metwalli et al., 2022) So, beyond transmitter class alone, the medial and extended amygdala are populated by several lineage-defined neuron groups whose different origins likely underlie some of the regional diversity in marker expression, connectivity, and function.

Evidence

(Vicario et al., 2014)

Genetic identification of the central nucleus and other components of the central extended amygdala in chicken during development

> *"Most of the subdivisions include various subpopulations of cells that apparently originate in the dorsal striatal, ventral striatal, pallidal, and preoptic embryonic domains, reaching their final location by either radial or tangential migrations."*

(Garcia-Calero et al., 2020)

Histogenetic Radial Models as Aids to Understanding Complex Brain Structures: The Amygdalar Radial Model as a Recent Example

> *"These results point to a heterogeneous cell type composition of various amygdalar regions, with at least pallial and subpallial subdivisions, and hypothalamic and various subpallial tangentially migrated cell types."*

(Vicario et al., 2016)

Genoarchitecture of the extended amygdala in zebra finch, and expression of FoxP2 in cell corridors of different genetic profile *"the medial amygdala also includes cell subpopulations from the same origins as those in the BSTM (pallidal, preoptic, hypothalamic, prethalamic eminence)"*

"In addition to these cells, the medial amygdala includes a minor subpopulation of Lhx9 cells of ventral pallial origin"

(Gerlach et al., 2021)

Neural pathways of olfactory kin imprinting and kin recognition in zebrafish

"the mammalian/rodent medial amygdala is a mosaic of GABAergic subpallial cells complemented by glutamatergic neuron types from extrinsic sources (ventral pallium, SPV, EmT)."

(Metwalli et al., 2022)

Distinct Subdivisions in the Transition Between Telencephalon and Hypothalamus Produce Otp and Sim1 Cells for the Extended Amygdala in Sauropsids

"TOH domain-derived double-labeled cells populated ventral parts of the medial extended amygdala, from medial to lateral levels (i.e., from BSTM3 to part of the Me) (Figures 10A,B, details in Figures 10D,D). SPV core-derived Sim1 single-labeled cells formed a distinct stripe of BSTM (Figure 10A, detail of single-labeled cells pointed with empty arrowheads in Figure 10E), and some were also scattered in other parts of the medial extended amygdala (Figures 10D,D). Sim1 cells also spread further dorsally reaching the arcopallium. However, in contrast to Otp cells, Sim1 cells reaching the arcopallium seem more abundant, and a subset of Sim1 cells also invades the central extended amygdala, mainly its lateral part (Figures 9D-F, 10C, detailed in Figure 10F)."

Non-neuronal and immature neuron populations

The medial temporal lobe and amygdala are not made only of mature neurons. They also contain large non-neuronal populations, especially glia, and in some medial temporal and amygdala regions they contain persistent immature neurons marked by proteins such as DCX, PSA-NCAM, Bcl-2, and sometimes NeuN. (8 sources)

A complete cell-type answer for the medial temporal lobe and amygdala has to add the non-neuronal cells that support, insulate, defend, and vascularize these circuits. At minimum, these include astrocytes, oligodendrocytes, oligodendrocyte precursor cells, microglia, and vascular-associated cells such as endothelial and mural cells. The references provided here especially emphasize that glia can make up a very large share of tissue composition in at least some amygdala subdivisions: one recent summary states that both left and right medial amygdala are composed primarily of glial cells, at over 70%, even though neurons still show hemispheric differences in number. (Ignacio et al., 2025)

The other major point is that some medial temporal and amygdalar zones contain long-lasting **immature neurons**, so the regional cell inventory includes more than just mature principal cells and interneurons. In the human temporal lobe near the paralaminar nucleus, Sorrells and colleagues found dense fields of **vimentin+ nestin+ cells** together with **DCX+ PSA-NCAM+ cells** in proliferative developmental zones, consistent with neural progenitor-like and immature neuronal populations linked to the caudal ganglionic eminence. (Sorrells et al., 2019) (Hansen et al., 2013) (Rubin et al., 2013) (Miyoshi et al., 2015)

In primates, these immature neuron populations are not restricted to one tiny amygdalar spot. Work summarized by Chareyron et al. argues that the immature neurons in the ventral amygdala belong to a broader stream of **Bcl-2-positive** and **NeuN-positive** cells extending from the lateral ventricle subventricular zone into the **paralaminar nucleus** and into nearby medial temporal cortical areas, including **anterior entorhinal cortex** and **layer II of perirhinal cortex**; these cells can also be seen with **DCX** labeling. (Chareyron et al., 2021) Recent review text makes the same point more broadly, noting adult mammalian **Bcl2+ immature neurons** in the **ventromedial amygdala** and **temporal cortex** of humans and non-human primates, with co-labeling for **β-tubulin-III, PSA-NCAM, DCX, and NeuN**, which supports the view that these are genuinely neuronal but still immature cells rather than only proliferating precursors. (Villard et al., 2023) (Sorrells et al., 2019) (Bernier et al., 1998)

So, for the user's cell-type question, the non-neuronal and immature-neuron part of the answer is: **glial cells** are a major population throughout medial temporal lobe and amygdala tissue; **progenitor-like cells** marked by nestin, vimentin, and Ki-67 can appear in developmental or niche-like zones; and some amygdalar and adjacent medial temporal regions contain **persistent immature neurons** marked by **DCX, PSA-NCAM, Bcl-2,**

β-tubulin-III, and NeuN, especially around the **paralaminar/ventral amygdala**, **entorhinal cortex**, and **perirhinal cortex**. (Ignacio et al., 2025) (Sorrells et al., 2019) (Chareyron et al., 2021) (Villard et al., 2023)

Evidence

(Ignacio et al., 2025)

The medial amygdala's neural circuitry: Insights into social processing and sex differences.

> *"Both the left and right medial amygdalae are primarily composed of glial cells (over 70 %). Still, the left amygdala contains 18 % more neurons than the right (Dall'Oglio, 2013)"*

(Sorrells et al., 2019)

Immature excitatory neurons develop during adolescence in the human amygdala

> *"The PL develops next to the CGE which is highly proliferative in humans (Hansen et al., 2013) . To determine whether the PL forms from progenitors in the CGE we immunostained the human temporal lobe for Ki-67 to label dividing cells, and for transcription factors expressed in the CGE (SP8, COUP-TFII, and PROX1) (Ma et al., 2013)(Hansen et al., 2013)(Miyoshi et al., 2015)(Rubin et al., 2013)"*

> *"Regions with high Ki-67 + cell density contained many SP8 + COUP-TFII + cells, few NKX2.1 + cells that would be likely be medial ganglionic eminence-derived, and a dense field of vimentin + nestin + cells and DCX + PSA-NCAM + cells (Fig. 1f, g)"*

(Hansen et al., 2013)

Non-epithelial stem cells and cortical interneuron production in the human ganglionic eminences

> *"GABAergic cortical interneurons underlie the complexity of neural circuits and are particularly numerous and diverse in humans. In rodents, cortical interneurons originate in the subpallial ganglionic eminences, but their developmental origins in humans are controversial. We characterized the developing human ganglionic eminences and found that the subventricular zone (SVZ) expanded massively during the early second trimester, becoming densely populated with neural stem cells and intermediate progenitor cells. In contrast with the cortex, most stem cells in the ganglionic eminence SVZ did not maintain radial fibers or orientation. The medial ganglionic eminence exhibited unique patterns of progenitor cell organization and clustering, and markers revealed that the caudal ganglionic eminence generated a greater proportion of cortical interneurons in humans than in rodents. On the basis of labeling of newborn neurons in slice culture and mapping of proliferating interneuron progenitors, we conclude that the vast majority of human cortical interneurons are produced in the ganglionic eminences, including an enormous contribution from non-epithelial SVZ stem cells."*

(Rubin et al., 2013)

PROX1: A Lineage Tracer for Cortical Interneurons Originating in the Lateral/Caudal Ganglionic Eminence and Preoptic Area

> *"The homeobox-encoding gene Prox1 and its Drosophila homologue prospero are key regulators of cell fate-specification. In the developing rodent cortex a sparse population of cells thought to correspond to late-generated cortical pyramidal neuron precursors expresses PROX1. Using a series of transgenic mice that mark cell lineages in the subcortical telencephalon and, more specifically, different populations of cortical interneurons, we demonstrate that neurons expressing PROX1 do not represent pyramidal neurons or their precursors but are instead subsets of cortical interneurons. These correspond to interneurons originating in the lateral/caudal ganglionic eminence (LGE/CGE) and a small number of preoptic area (POA)-derived neurons. Expression within the cortex can be detected from late embryonic stages onwards when cortical interneurons are still migrating. There is persistent expression in postmitotic cells in the mature brain mainly in the outer cortical layers. PROX1+ve interneurons express neurochemical markers such as calretinin, neuropeptide Y, reelin and vasoactive intestinal peptide, all of which are enriched in LGE/CGE- and some POA-derived cells. Unlike in the cortex, in the striatum PROX1 marks nearly all interneurons regardless of their origin. Weak expression of PROX1 can also be detected in oligodendrocyte lineage cells throughout the forebrain. Our data show that PROX1 can be used as a genetic lineage tracer of nearly all LGE/CGE- and subsets POA-derived cortical interneurons at all developmental and postnatal stages in vivo."*

(Miyoshi et al., 2015)

Prox1 Regulates the Subtype-Specific Development of Caudal Ganglionic Eminence-Derived GABAergic Cortical Interneurons

> *"Neurogliaform (RELN+) and bipolar (VIP+) GABAergic interneurons of the mammalian cerebral cortex provide critical inhibition locally within the superficial layers. While these subtypes are known to originate from the embryonic caudal ganglionic eminence (CGE), the specific genetic programs that direct their positioning, maturation, and integration into the cortical network have not been elucidated. Here, we report that in mice expression of the transcription factor Prox1 is selectively maintained in postmitotic CGE-derived cortical interneuron precursors and that loss of Prox1 impairs the integration of these cells into superficial layers. Moreover, Prox1 differentially regulates the postnatal maturation of each specific subtype originating from the CGE (RELN, Calb2/VIP, and VIP). Interestingly, Prox1 promotes the maturation of CGE-derived interneuron subtypes through intrinsic differentiation programs that*

operate in tandem with extrinsically driven neuronal activity-dependent pathways. Thus Prox1 represents the first identified transcription factor specifically required for the embryonic and postnatal acquisition of CGE-derived cortical interneuron properties. SIGNIFICANCE STATEMENT Despite the recognition that 30% of GABAergic cortical interneurons originate from the caudal ganglionic eminence (CGE), to date, a specific transcriptional program that selectively regulates the development of these populations has not yet been identified. Moreover, while CGE-derived interneurons display unique patterns of tangential and radial migration and preferentially populate the superficial layers of the cortex, identification of a molecular program that controls these events is lacking. Here, we demonstrate that the homeodomain transcription factor Prox1 is expressed in postmitotic CGE-derived cortical interneuron precursors and is maintained into adulthood. We found that Prox1 function is differentially required during both embryonic and postnatal stages of development to direct the migration, differentiation, circuit integration, and maintenance programs within distinct subtypes of CGE-derived interneurons."

(Chareyron et al., 2021)

Life and Death of Immature Neurons in the Juvenile and Adult Primate Amygdala

"Our current findings demonstrate that the population of immature neurons found in the ventral amygdala is not an isolated population but may be part of a larger group of Bcl-2-positive and NeuN-positive cells that extends from the SVZ of the lateral ventricle to the paralaminar nucleus and some cortical areas within the medial temporal lobe. The presence of these immature neurons in the ventral amygdala, anterior entorhinal cortex and layer II of the perirhinal cortex can be observed on coronal or sagittal sections of the temporal lobe labeled with DCX or Bcl-2 markers (Bernier et al., 2002)(Fudge, 2004)(0)."

(Villard et al., 2023)

Structural plasticity in the entorhinal and perirhinal cortices following hippocampal lesions in rhesus monkeys

"populations of immature neurons expressing the anti-apoptosis Bcl2 protein have been observed in several regions of the adult mammalian brain, including the ventromedial amygdala and the temporal cortex in humans and non-human primates (Bernier et al., 2002)(Bernier et al., 1998)(Fudge, 2004)(Yachnis et al., 2000)"

".These cells are immunoreactive for other markers of neuronal immaturity, such as β-tubulin-III, PSA-NCAM and doublecortin (DCX) (Bernier et al., 2002)(Chareyron et al., 2016)(Fudge, 2004)(Yachnis et al., 2000), and also express the neuronal nuclear antigen (NeuN), thus providing evidence that these cells are neurons (Chareyron et al., 2016)(Sorrells et al., 2019)(Yachnis et al., 2000)(0)."

(Bernier et al., 1998)

Bcl-2 Protein as a Marker of Neuronal Immaturity in Postnatal Primate Brain

"The distribution of neurons expressing immunoreactivity for the protein Bcl-2 was studied in the brain of squirrel monkeys (Saimiri sciureus) of various ages. Several subsets of small and intensely immunoreactive neurons displaying an immature appearance were disclosed in the amygdala and piriform cortex. The piriform cortex exhibited clusters of various forms in which Bcl-2+ neurons appeared linked to one another by their own neurites. The subventricular zone, which is known to harbor the largest population of rapidly and constitutively proliferating cells in the adult rat brain, was intensely stained, particularly at the basis of the lateral ventricle. A long and dorsoventrally oriented Bcl-2+ fiber fascicle was seen to emerge from the subventricular zone, together with numerous Bcl-2+ cells that formed a densely packed column directed at the olfactory tubercle. In adult and aged monkeys, the small and intensely labeled neurons were progressively replaced by larger and more weakly stained neurons in the amygdala and piriform cortex. In contrast, Bcl-2 immunostaining did not change with age in the subventricular zone and olfactory tubercle, the islands of Calleja of which were markedly enriched with Bcl-2. The dentate gyrus contained only a few layers of intensely labeled granule cells in juvenile monkeys, but the number of these layers increased markedly in adult and aged monkeys. These findings suggest that Bcl-2 can serve as a marker of both proliferating and differentiating neurons and indicate that such immature neurons may be much more widespread than previously thought in postnatal primate brain."

Cell-type diversity maps and specialized functional neuron classes

New atlas-style studies show that cell diversity in the medial temporal lobe is even larger than older anatomy-based lists suggested. They also show that some neurons are best defined by what they encode, such as fear, extinction, concepts, or spatial location, not just by marker genes or shape. (3 sources)

Recent cell atlas work makes it clear that any answer to "all the cell types" in the medial temporal lobe and amygdala has to stay open-ended, because large-scale single-cell mapping can split broad classes into many finer neuronal types and states. In one recent mouse hippocampal formation atlas, investigators identified **130 neuronal cell types** and mapped where they sit in space, showing how much hidden diversity exists even within a single major medial temporal lobe system. (Hochgerner et al., 2023) This kind of result supports the idea that the medial temporal lobe and amygdala are populated not only by the broad classes already listed above, but by many transcriptomic subtypes nested inside those classes.

A second lesson is that some important neuron populations are defined functionally. In the basal amygdala, work on emotional learning has highlighted specialized **fear neurons** and **extinction neurons**, showing that even within one amygdala nucleus there are functionally distinct ensembles tied to opposite learned states. (Carrere et al., 2015) In human medial temporal lobe recordings, a different kind of specialization appears: **concept cells** have been described in the **hippocampus, amygdala, and entorhinal cortex**, while **parahippocampal location-selective neurons** form another specialized class linked to spatial coding. (Mackay et al., 2024) So the most complete answer for the user is that the medial temporal lobe and amygdala contain not just region-based excitatory, inhibitory, glial, immature, and lineage-defined populations, but also many finer transcriptomic cell types and specialized functional neuron classes whose identity depends on what they represent in behavior and memory.

Evidence

(Hochgerner et al., 2023)

Neuronal types in the mouse amygdala and their transcriptional response to fear conditioning

> *"identify 130 neuronal cell types and validate their spatial distributions."*

(Carrere et al., 2015)

A pavlovian model of the amygdala and its influence within the medial temporal lobe

> *"Recent advances in neuroscience give us a better view of the inner structure of the amygdala, of its relations with other regions in the Medial Temporal Lobe (MTL) and of the prominent role of neuromodulation. They have particularly shed light on two kinds of neurons in the basal nucleus of the amygdala, the so-called fear neurons and extinction neurons."*

(Mackay et al., 2024)

Concept and location neurons in the human brain provide the ‘what’ and ‘where’ in memory formation

> *"we could attribute such effects to two specialized sub-groups of neurons: concept cells in the hippocampus, amygdala and entorhinal cortex (EC), and a second group of parahippocampal location-selective neurons."*